HW 0317

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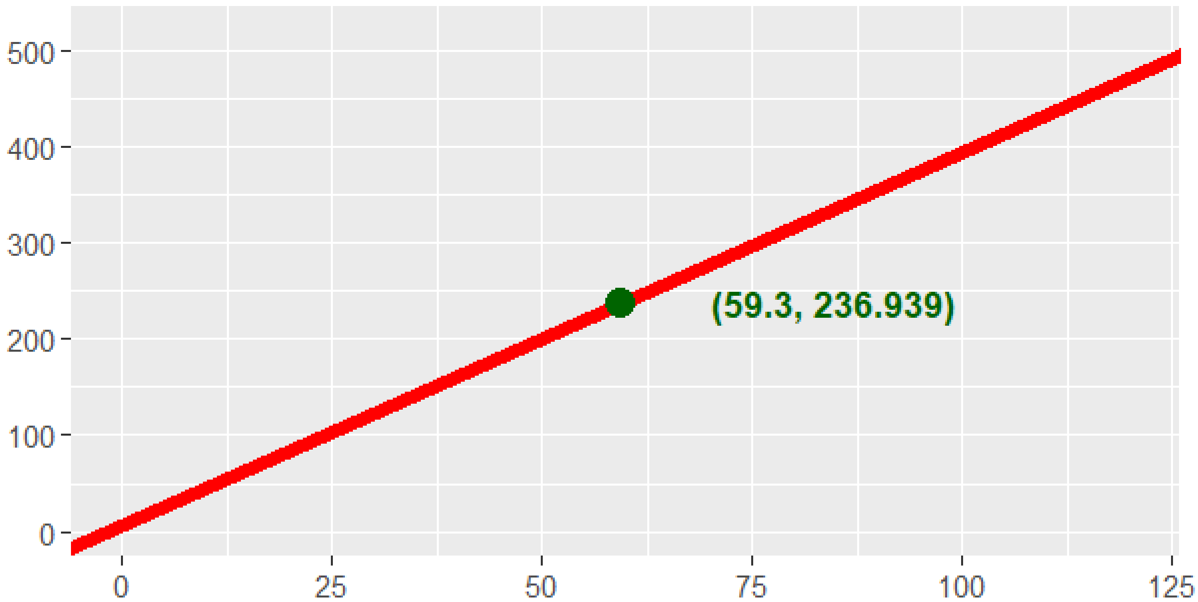
18.

(a)

$$\widehat{INSURANCE} = \underset{(7.383)}{6.855} + \underset{(0.112)}{3.880} \times INCOME$$

$$\begin{aligned}&\text{slope : } 3.880, \text{intercept : } 6.855\\&\overline{\text{INCOME}} = 59.3,\\&\overline{\text{INSURANCE}}\\&= 6.855 + 3.880 * \overline{\text{INCOME}}\\&= 6.855 + 3.880 * 59.3 = 236.939\end{aligned}$$



(b)

Since $\hat{\beta}_2 = 3.880$, for every $1,000 increase in household income, the estimated average amount of life insurance held increases by **$3,880**

$$\begin{aligned}df &= N - 2 = 20 - 2 = 18\\t_c &= t_{0.05/2,18} \approx 2.101\end{aligned}$$

$$\begin{aligned}\text{Point Estimate} &= 3.880,\\95\ \%\ \text{Interval Estimate} &= [\hat{\beta}_2 - (t_c * se(\hat{\beta}_2)), \hat{\beta}_2 + (t_c * se(\hat{\beta}_2))]\\&= [3.880 - (2.101 * 0.112), 3.880 + (2.101 * 0.112)]\\&= [3.6447, 4.1153]\end{aligned}$$

In repeated sampling, about 95% intervals constructed this way will contain the true value of the parameter β_2.

(c) Construct a 99% interval estimate of $E(y|x_0 = 100)$
Note that $cov(b_1, b_2) = -0.746, t_c = t_{0.01/2,18} \approx 2.878$

$$\begin{aligned}
\hat{y}_0 &= b_1 + b_2(100) = 6.855 + 3.880(100) = 394.855 \\
var(\hat{y}_0) &= var(b_1) + (100)^2 var(b_2) + 2(100)cov(b_1, b_2) \\
&= 7.383^2 + 10000 * 0.112^2 + 200 * (-0.746) \\
&= 30.7487 \\
se(\hat{y}_0) &= \sqrt{var(\hat{y}_0)} = \sqrt{30.7487} = 5.5452
\end{aligned}$$

$$\begin{aligned}
\text{99 \% Interval Estimate} &= [\hat{y}_0 - (t_c * se(\hat{y}_0)), \hat{y}_0 + (t_c * se(\hat{y}_0))] \\
&= [394.855 - (2.878 * 5.5452), 394.855 + (2.878 * 5.5452)] \\
&= [378.896, 410.814]
\end{aligned}$$

(d)

$$\text{Null Hypothesis } H_0 : \beta_2 = 5$$
$$\text{Alternative Hypothesis } H_1 : \beta_2 \neq 5$$

Using t-statistic:

$$t^* = \frac{b_2 - 5}{se(b_2)} = \frac{3.880 - 5}{0.112} = -10$$

Rejection Region:

$$\{t \mid |t| > t_{0.025,18} \approx 2.101\}$$

Since $t^* \in \{t \mid |t| > t_{0.025,18} \approx 2.101\}$, reject Null Hypothesis H_0 at the 5% significance level

(e)

$$\text{Null Hypothesis } H_0 : \beta_2 = 1$$
$$\text{Alternative Hypothesis } H_1 : \beta_2 > 1$$

Using t-statistic:

$$t^* = \frac{b_2 - 1}{se(b_2)} = \frac{3.880 - 1}{0.112} = 25.714$$

Rejection Region:

$$\{t \mid t > t_{0.01,18} \approx 2.552\}$$

Since $t^* \in \{t \mid t > t_{0.01,18} \approx 2.552\}$, reject Null Hypothesis H_0 at the 1% significance level

Conclude that as income increases, the amount of life insurance held increases by significantly more than the increase in income.

23.
(a)

$$\begin{aligned}
\widehat{PRICE} &= \underset{(6.0722)}{93.5659} + \underset{(0.0053)}{0.1845} \times SQFT^2 \\
N &= 500, \quad R^2 = 0.7122
\end{aligned}$$

$$\frac{dPRICE}{dSQFT} = \frac{d}{dSQFT}(\alpha_1 + \alpha_2 SQFT^2) = 2\alpha_2 SQFT$$

Marginal effect of a 2000 square foot house $= 2\alpha_2 SQFT|_{SQFT=20} = 40\alpha_2$

$$\text{Null Hypothesis } H_0 : 40\alpha_2 \leq 13$$
$$\text{Alternative Hypothesis } H_1 : 40\alpha_2 > 13$$

Using t-statistic:

$$t^* = \frac{40b_2 - 13}{se(40b_2)} = \frac{40 * 0.1845 - 13}{40 * 0.0053} = -26.7278$$

Rejection Region:

$$\{t \mid t > t_{0.05,500-2} \approx 1.648\}$$

p-value:

$$P(T > t^*) = 1$$

Since p-value : $1 > 0.05$, we do NOT reject H_0. For a 2000 square foot house, there's no evidence that marginal effect will be greater than \$13000.

(b)
Marginal effect of a 2000 square foot house $= 2\alpha_2 SQFT|_{SQFT=40} = 80\alpha_2$

$$\text{Null Hypothesis } H_0 : 80\alpha_2 \leq 13$$
$$\text{Alternative Hypothesis } H_1 : 80\alpha_2 > 13$$

Using t-statistic:

$$t^* = \frac{80b_2 - 13}{se(80b_2)} = \frac{80 * 0.1845 - 13}{80 * 0.0053} = 4.1893$$

Rejection Region:

$$\{t \mid t > t_{0.05,500-2} \approx 1.648\}$$

p-value:

$$P(T > t^*) = 1.64e - 5$$

Since p-value : $1.64e - 5 < 0.05$, we reject H_0. For a 4000 square foot house, the marginal effect of increasing the size by 100 square feet is significantly greater than \$13,000.

(c)

$$\begin{aligned} E(PRICE \mid SQFT = 20) &= \hat{\alpha}_1 + \hat{\alpha}_2(20^2) \\ &= 93.5659 + 0.184519 \times 400 \\ &= 167.3735 \end{aligned}$$

For standard error:

$$\begin{aligned} se(PRICE \mid SQFT = 20) &= \sqrt{var(PRICE \mid SQFT = 20)} \\ &= \sqrt{var(\hat{\alpha}_1) + (400)^2 var(\hat{\alpha}_2) + 2(400)cov(\hat{\alpha}_1, \hat{\alpha}_2)} \\ &= \sqrt{36.872 + 160000 * 0.00002762 + 800 * (-0.023)} \\ &= \sqrt{22.891} = 4.768 \end{aligned}$$

$$t_c = t_{0.05/2,500-2} \approx 1.965$$

Similar to 18.(b)

$$\begin{aligned} 95\ \%\ \text{Interval Estimate} &= [167.3735 - (1.965 * 4.768), 167.3735 + (1.965 * 4.768)] \\ &= [158.0044, 176.7426] \end{aligned}$$

In repeated sampling, about 95% intervals constructed this way will contain the true value of the parameter β_2.

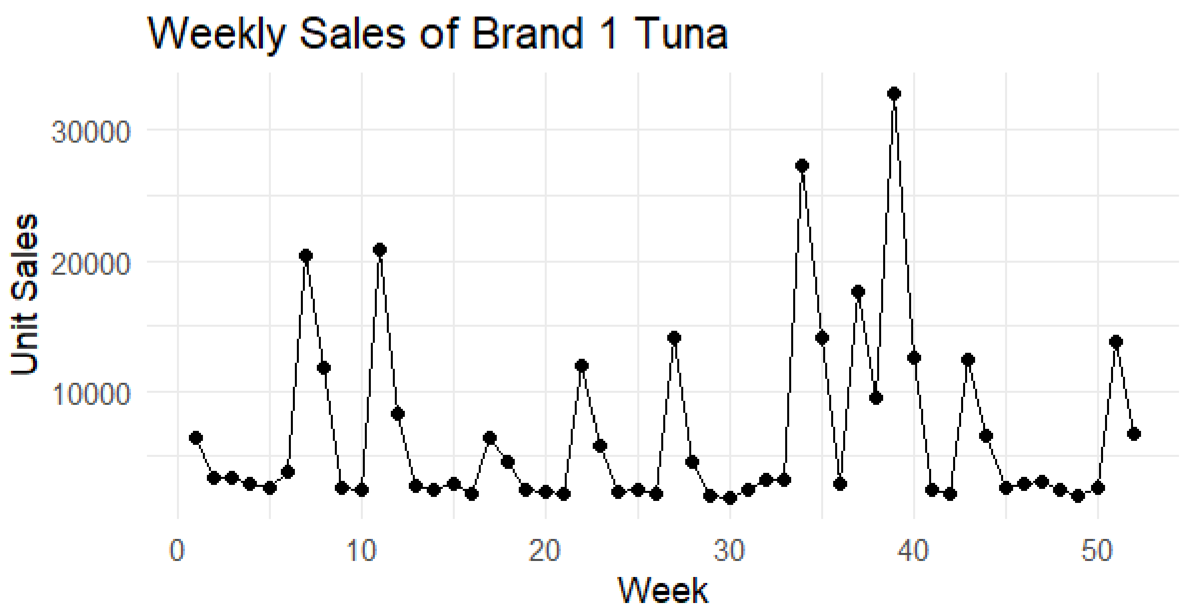
(d)

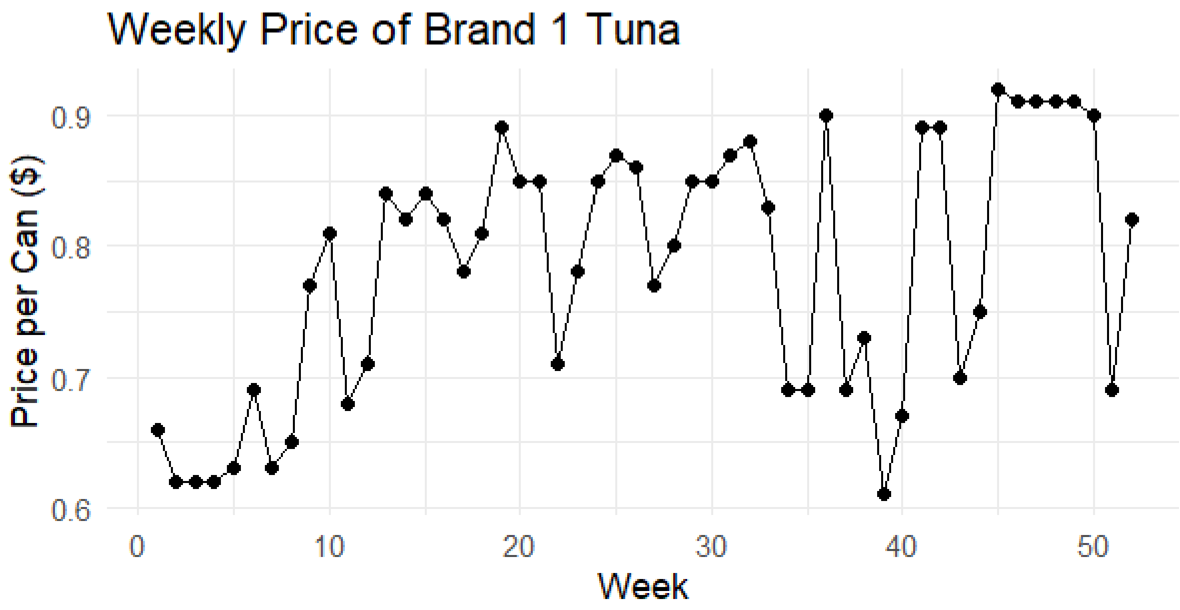
There are **3** houses with a living area of exactly 2,000 square feet. Sample mean = 163.33. The sample average of 163.33 is compatible with the results in part (c) because it falls well within the 95% confidence interval of [157.92, 176.83].

31.

(a)

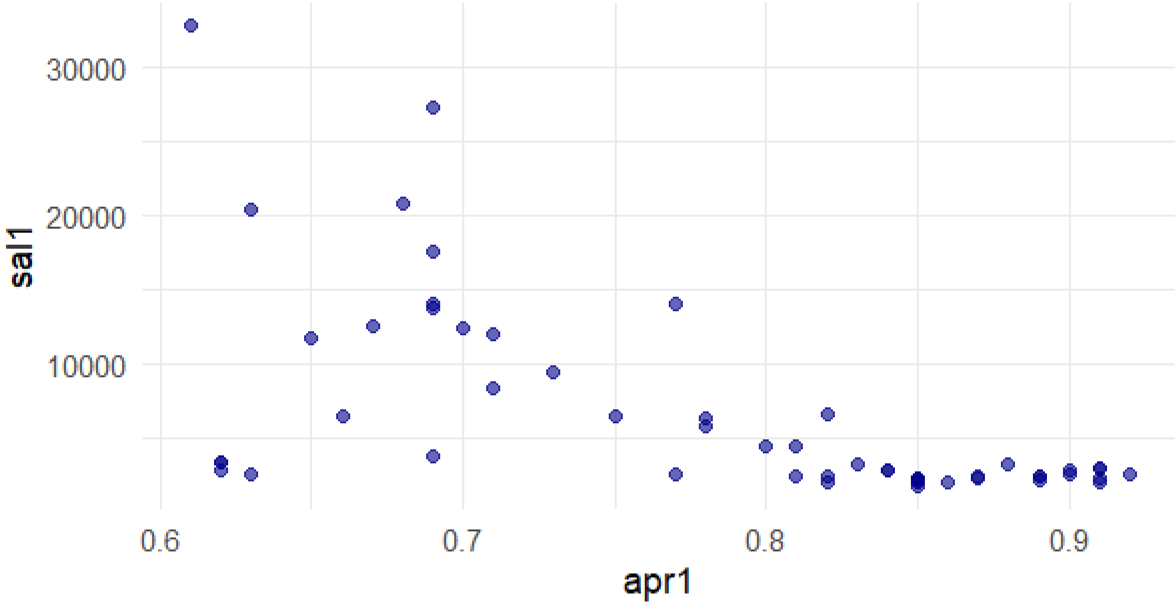
Variable	Mean	Std. Dev.	Min	Max
SAL1 (Unit Sales)	6719	6915.083	1762	32820
APR1 (Price per Can)	0.7825	0.098	0.6100	0.9200





SAL1: The variation in sales is extremely high.The data shows massive spikes in specific weeks. APR1: The price shows discrete variation. Instead of changing every week, it remains constant for several periods before dropping significantly .

(b)



It seems to be a negative relationship between sales and price. This is what is expected since the result is consistent with the fundamental Law of Demand in economics (Inverse relationship between the price of a good and the quantity demanded.)

(c)

$$\widehat{SAL1} = \underset{(6196.42)}{40714.22} - \underset{(78.58)}{434.45} \times PRICE1$$
$$N = 52, \quad R^2 = 0.3794$$

The coefficient of point estimate is -434.45. This means that the effect of an one-cent increase in the price of brand no. 1 is -434.45.

$$95\% \text{ confidence interval} : [-592.29, -276.60]$$

(d)

$$90\% \text{ Confidence Interval: } [8624.934, 11980.87]$$

(e)

$$\hat{\epsilon} = \hat{\beta}_2 \times \frac{\bar{X}}{\bar{Y}} = -434.45 \times \frac{78.25}{6718.712} = -5.059856$$

$$se(\hat{\epsilon}) = se(\hat{\beta}_2) \times \frac{\bar{X}}{\bar{Y}} = 78.58 \times \frac{78.25}{6718.712} \approx 0.9151881$$

$$\begin{aligned} 95\ \% \text{ Interval Estimate} &= [\hat{\epsilon} - t_{0.05/2,50-2} * se(\hat{\epsilon}), \hat{\epsilon} + t_{0.05/2,50-2} * se(\hat{\epsilon})] \\ &= [-5.059856 - (2.010635 * 0.9151881), -5.059856 + (2.010635 * 0.9151881)] \\ &= [-6.899965, -3.219747] \end{aligned}$$

Since the absolute value of the elasticity $|\epsilon| = 5.059856 > 1$, Even at the upper bound of the 95% confidence interval ($|-3.219747| > 1$), demonstrating that the product exhibits significant price elasticity at the means.
It makes economic sense since tuna are not unique or essential. When a specific brand of tuna increases its price, it is not a "necessity" that consumers must buy at any cost. Thus the price could change elasticity.

(f)

$$\epsilon = \beta_2 \frac{\bar{X}}{\bar{Y}}$$

$$\text{Null Hypothesis } H_0 : \epsilon = -3$$
$$\text{Alternative Hypothesis } H_1 : \epsilon \neq -3$$

Using t-statistic:

$$t^* = \frac{\beta_2 \frac{\bar{X}}{\bar{Y}} - (-3)}{se(\beta_2 \frac{\bar{X}}{\bar{Y}})} = \frac{-5.059856 + 3}{0.9151881} = -2.251$$

Rejection Region:

$$\{t \mid |t| > t_{0.05,48} \approx 1.677224\}$$

p-value:

$$2 \times P(T > |t^*|) = 0.029$$

Since $t^* \in \{t \mid |t| > t_{0.05,48} \approx 1.677224\}$, reject Null Hypothesis H_0 at the 10% significance level

Conclude that the price elasticity of sales for brand no. 1 is significantly different from -3.